



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

CHI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on lightweight routing, middleware, context, and net/http

TOTAL: 10 QUESTIONS

Q1 What is Chi and why is it described as idiomatic Go?

Threat Model

Q6 How do you implement JWT authentication middleware in Chi?

Observability

Q2 How does Chi middleware chaining work?

Architecture

Q7 What built-in middleware does Chi provide?

Lifecycle

Q3 How do you read URL parameters in Chi?

Configuration

Q8 How do you define RESTful resource routes in Chi?

Compliance

Q4 What is `chi.Mount()` and how do you use it?

Hardening

Q9 How does Chi handle 404 and 405 responses?

Testing

Q5 How do you pass values between middleware and handlers in Chi?

SSO / IdP

Q10 How do you write tests for Chi handlers?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Chi is a lightweight router that

Mitigation

Chi is a lightweight router that implements `http.Handler` and uses only standard `net/http` no custom context types. Middleware is standard `func(http.Handler) http.Handler`. Any existing `net/http` handler works out of the box without adaptation.

A6 Write a `func(next http.Handler) http.Handler`

Observability

Write a `func(next http.Handler) http.Handler` that reads the `Authorization` header, validates the JWT, and on success calls `next.ServeHTTP(w, r.WithContext(ctx))` with the user in the context. On failure, write 401 and return without calling next.

A2 Chi uses alice-style functional composition.

Primitives

Chi uses alice-style functional composition. `r.Use(middleware.Logger)` applies globally. `r.Group(func(r chi.Router) { r.Use(AuthMiddleware); r.Get('/admin', handler) })` scopes middleware to the group. Each middleware wraps the next `http.Handler`.

A7 `middleware.Logger` (structured request logging),

Automation

`middleware.Logger` (structured request logging), `middleware.Recoverer` (panic recovery), `middleware.Throttle(N)` (concurrent request limiter), `middleware.Compress` (gzip), `middleware.StripSlashes`, `middleware.Heartbeat('/health')`, and `middleware.RealIP`.

A3 `chi.URLParam(r, 'id')` reads the named parameter

Hardening

`chi.URLParam(r, 'id')` reads the named parameter from the URL. In a handler: `id := chi.URLParam(r, 'id')`. Chi stores params in the request context under its own key, making them accessible without a custom `Context` type via the standard library.

A8 `r.Get('/users', listUsers); r.Post('/users',`

Governance

`r.Get('/users', listUsers); r.Post('/users', createUser); r.Route('/users/{id}', func(r chi.Router) { r.Get('/', getUser); r.Put('/', updateUser); r.Delete('/', deleteUser) })`. Chi does not have a `resources()` shortcut explicit routes are idiomatic.

A4 `r.Mount('/api', apiRouter)` attaches a sub-router at

Gotchas

`r.Mount('/api', apiRouter)` attaches a sub-router at a prefix. `apiRouter` is itself a `*chi.Mux`. `Mount` strips the prefix before passing the request to the sub-router. This enables fully independent route trees for modular codebases.

A9 `r.NotFound(handler)` sets a custom 404 handler

Assessment

`r.NotFound(handler)` sets a custom 404 handler for unmatched routes. `r.MethodNotAllowed(handler)` sets a 405 handler when a route exists but not for the used HTTP method. Without custom handlers, Chi returns plain text 404/405 by default.

A5 Use standard context: `ctx :=`

Federation

Use standard context: `ctx := context.WithValue(r.Context(), userKey{}, user); r = r.WithContext(ctx)`. Retrieve in the handler: `user := r.Context().Value(userKey{}).(User)`. Chi recommends unexported struct keys to avoid collisions with other packages.

A10 Chi handlers are plain `http.HandlerFunc`, so

Optimization

Chi handlers are plain `http.HandlerFunc`, so use `httptest.NewRequest` and `httptest.NewRecorder` directly. Set chi URL params with `chitest: rctx := chi.NewRouteContext(); rctx.URLParams.Add('id', '42'); r = r.WithContext(context.WithValue(r.Context(), chi.RouteCtxKey, rctx))`.